Juliette Grosset

Adelaide, South Australia, Australia | jgrosset10@gmail.com | grossetjuliette | juliette-grosset | Google Scholar

Experience

Naval Group / IRL CROSSING, CNRS, Post-Doctoral Researcher — Adelaide, Australia, Feb 2025 – present

- Development of **LOTUSim**, an open-source real-time maritime simulation platform for human-vehicle teaming, and of **LOTUSim-Energy**, its energy-oriented extension developed within the FACET project.
- Real-time physics-based ocean current simulation and multi-layer underwater current modelling for maritime robotics.
- Benchmarking of Gazebo-based and alternative maritime simulation environments for swarm robotics and AI training.
- Member of the LOTUSim Technical Committee and co-leader of the Maritime Working Group at IRL CROSSING; in 2026, Naval Group representative on the IRL CROSSING Management Committee, and elected as a non-permanent member of the IRL CROSSING Laboratory Council.

IMT Atlantique / IRISA, Research Engineer — Rennes, France, 2024 – 2024

- Simulation of AGV fleets with the CARLA simulator and V2X cooperation experiments.

ECAM Rennes - Louis de Broglie / IMT Atlantique / IRISA, PhD Student — Rennes, France, 2021 – 2024

- Shared-perception-based obstacle and collision avoidance strategy for AIV fleets.
- Adaptation of V2X communication standards to Industry 4.0 and a V2X data generation architecture.
- Decentralised dynamic task (re)allocation mechanism and fuzzy-model-based collective energy management.

IMT Atlantique, Lab-STICC, Laboratory Technician — Brest, France, 2021 – 2021

IMT Atlantique, Lab-STICC, Research Intern — Brest, France, 2021 – 2021

ENIB, Pedagogical Tutor — Brest, France, 2021 – 2021

INRIA / IRISA, Assistant Engineer Intern — Rennes, France, 2020 – 2020

King Mongkut's University of Technology Thonburi (KMUTT), Technical Intern — Bangkok, Thailand, 2019 – 2019

Education

IMT Atlantique, Computer Science — Rennes, France, 2021 – 2024

- Thesis: *Collective intelligence strategies for efficient autonomous industrial vehicles*, prepared at IMT Atlantique and ECAM Rennes - Louis de Broglie (EASE team, IRISA).
- PhD supervisors: Jean-Marie Bonnin (Professor, IMT Atlantique) and Alain-Jérôme Fougères (Professor, ECAM Louis de Broglie). Co-supervision: Moïse Djoko-Kouam (ECAM Louis de Broglie).

- Reviewers: Leila Merghem-Boulahia and Gérard Chalhoub. Jury: Maroua Bouzid (chair), Leila Merghem-Boulahia, Gérard Chalhoub, Cédric Buche.
- Awarded **Best Thesis of the SPIN Doctoral School** (2025). Co-funded by Région Bretagne (ARED grant) and ECAM Louis de Broglie.

UBO / ENSTA Bretagne / IMT Atlantique / ENIB, Intelligent, Interactive and Autonomous Systems (SIIA) — Brest, France, 2020 – 2021

ENIB - École Nationale d'Ingénieurs de Brest, Computer Science, Electronics and Mechatronics — Brest, France, 2018 – 2021

- Specialised modules in Artificial Intelligence, Simulation and Robotics.

Université de Sherbrooke, Engineering — Sherbrooke, Canada, 2018 – 2018

Awards

Best Thesis of the SPIN Doctoral School — 2025
SPIN Doctoral School
ed-spin.doctorat-bretagne.fr/fr/actualites/prix-de-these-2025-decerne-juliette-grosset

Best Paper Award — 2024
ASPAI 2024

Best Presentation Award (ex aequo) — 2024
Colloque IMT 2024

4th Prize - Ma Thèse en 180 secondes — 2024
Ma Thèse en 180 secondes, Rennes qualification round

Best Poster Award (ex aequo) — 2022
Colloque IMT 2022

Research Interests

Keywords: Distributed systems, robotics, artificial intelligence, simulation, cooperative autonomy, maritime context, ITS context, human-in-the-loop.

Grants

2026-2027: [Member] LOTUSim-Energy - FACET (French and Australian Government funding). Maritime and energy. Approx. 135 k€.

2025: [Member] LOTUSim-Energy - FACET, in collaboration with Reuniwatt. Maritime and energy. Approx. 50 k AUD.

Leadership and Committees

2026 - present: Elected, Non-permanent member, IRL CROSSING Laboratory Council.

2026: Naval Group representative, IRL CROSSING Management Committee.

2026 - present: Co-leader, Maritime Working Group, IRL CROSSING (Australia).

2025 - present: Member, LOTUSim Technical Committee (Naval Group).

2025: Lead of the LOTUSim open-source release - the first open-source simulator in Naval Group's history.

Service

Reviewing (conferences): IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.

Reviewing (journals): SN Computer Science (Springer Nature, 2026); Journal of Autonomous Intelligence (2023).

Hiring committees: Research Engineer positions at IRL CROSSING / CNRS (2026) and Naval Group / IRL CROSSING (2025); Post-Doctoral position at ECAM Rennes - Louis de Broglie (2023).

Event organisation: Co-organiser of the FACET Offshore Wind Maintenance Workshop, Adelaide (2025); organising committee of PFIA 2022 and JJCR 2022 (France).

Scientific groups: Maritime Working Group, IRL CROSSING (2026-); GDR Robotique (2022); French Artificial Intelligence Association - AFIA (2022).

Student Supervision

Loris Cazaux 2026

- Towards an abstract and modular architecture for LOTUSim-Energy.

Baptiste Piar 2026

- Building a controller baseline for AUVs — from PID towards MPC under flowing currents.

Ella James 2026

- Development and comparison of wind models for simulating wind turbine energy output.

Riley Morrison 2026

- Migration of vision-based SLAM pipeline to ROS2 for lunar rover navigation, validation and deployment.

Kartikeya Dashora 2026

- LOTUSim — navigation control integration and experimental validation.

Hayato Donoghue 2026

- LOTUSim development and enhancement through game design, simulation content creation and environment development. Focus on simulator experience, visual design, scene refinement, training scenarios, realism, user experience, visual consistency and optimisation.

Bevan Shajan 2026

- Integration of crack detection algorithms into LOTUSim, exploration of event cameras for crack and corrosion detection, comparison of detection approaches through simulation and potential real-world experiments, and enhancement of LOTUSim for maritime inspection research.

Gechhai Thornthea 2026

- Integration of Gymnasium into LOTUSim.

Adelaide University project group 2026

- Integration of Pixhawk-based drone systems into LOTUSim using ROS 2 and Gazebo, enabling manual control and mission-based simulation.

Juan Cuevas 2024

- Real-world robot experimentation of a cooperative intersection algorithm using ROS 2 and V2X communication.

Kazuto Matsumoto 2024

- Deep learning approaches for predicting obstacle position and velocity from virtual camera inputs in the CARLA simulator.

Satoru Furuta 2024

- Camera-based obstacle detection in Gazebo and BIM-oriented environment reconstruction.

Iheb Hamed 2024

- Study and development of a multi-task mission management solution based on ground drones in a ROS 2 environment.

IMT Atlantique project group 2023-2024

- Simulation of vehicle–infrastructure interactions with the CARLA simulator, including industrial robots in a simplified urban environment of Rennes.

Rihab Binous 2023

- Study and development of a multi-task mission management solution performed by heterogeneous mobile robots.

ECAM Rennes project group 2022-2023

- Improvement of the performance and communication of a drone fleet.

Evaluation Juries

2025 (Sep.): 1 engineering internship defence, ENSTA.

2023 (Oct.): 6 foreign-internship defences (in English) and 1 technical internship defence, ECAM Rennes.

2022 (Oct.): 3 engineering-degree defences, ECAM Rennes.

2022 (Jan.): 1 project defence, ECAM Rennes (3rd year, Terrestrial Drones module).

Invited Talks

2025: SPIN Doctoral School Welcome Days - PhD Prize ceremony.

2024: Jeudi de la Recherche 2024, ECAM Rennes - Louis de Broglie.

2023: Jeudi de la Recherche 2023, ECAM Rennes - Louis de Broglie.

International Mobility

2025-2026: Adelaide, Australia (2 years) - Post-Doctoral Researcher, IRL CROSSING / Naval Group.

2023: Tokyo, Japan (3 months) - PhD research mobility, Tsukada Laboratory, University of Tokyo.

2019: Bangkok, Thailand (3 months) - research internship, KMUTT.

2018: Sherbrooke, Canada (4 months) - academic exchange semester.

Science Communication

- **Ma Thèse en 180 secondes** (2024): 4th jury prize at the Rennes qualification round and participation in the regional final in Brest.
- **L codent L créent** (2023): 8 sessions of Python programming workshops for middle-school girls in Rennes, addressing gender stereotypes in science and technology.

Open Source Projects

LOTUSim

LOTUSim-Energy

LOTUSim-generic-scenario

LOTUSim-Unity-modules

AIV Simulator

CARLA-ROS2-OpenCDA architecture

Languages

French

English